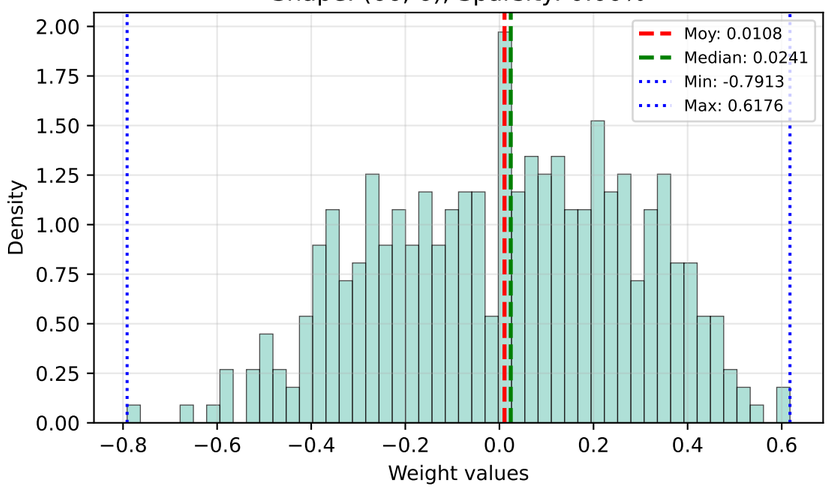
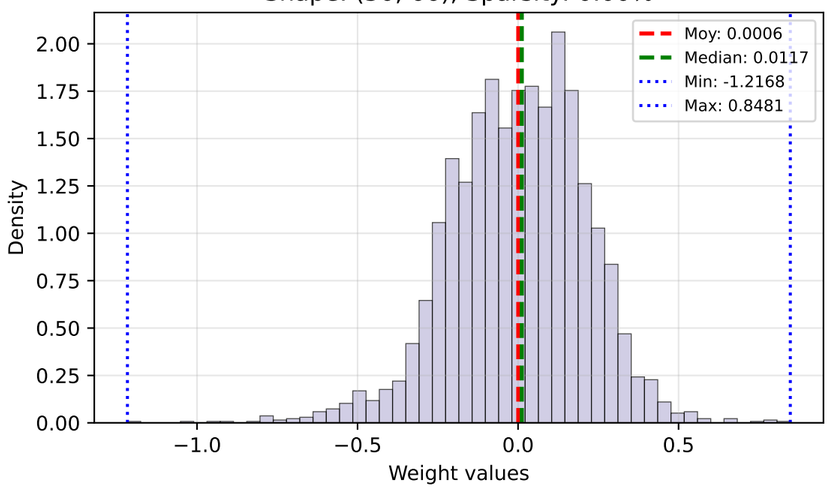
Detailed per-layer distributions - Weights and Biases

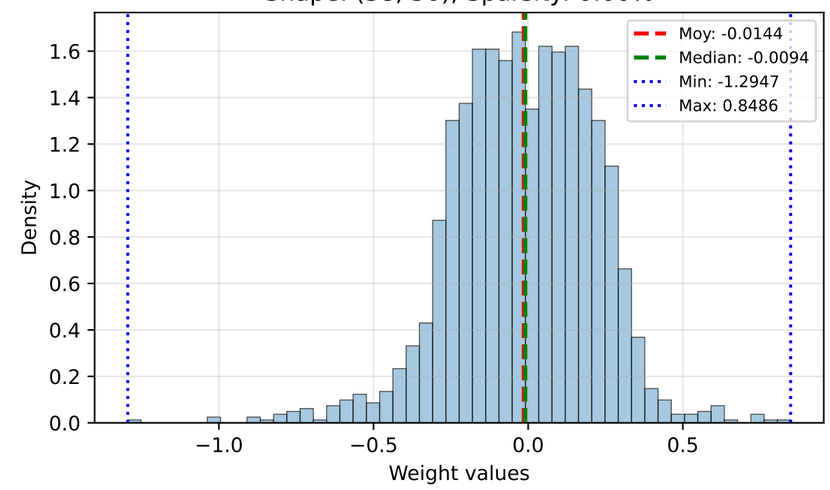
Layer 0 - WEIGHTS
Shape: (66, 6), Sparsity: 0.00%



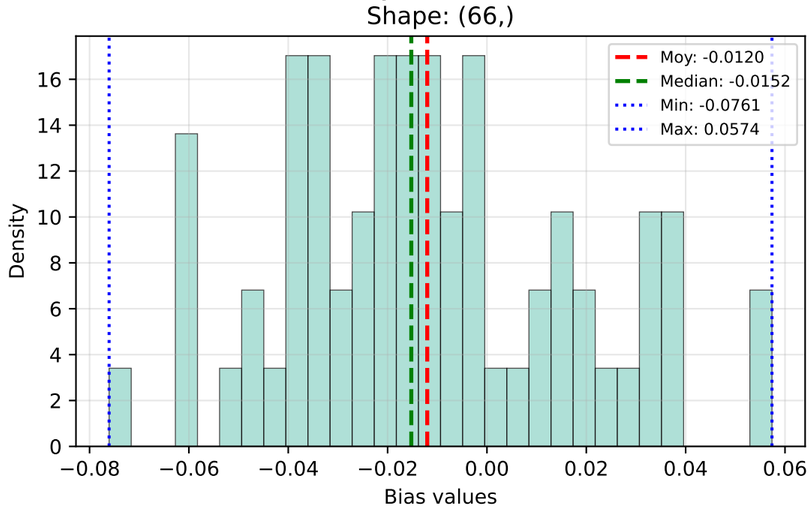
Layer 1 - WEIGHTS
Shape: (50, 66), Sparsity: 0.00%



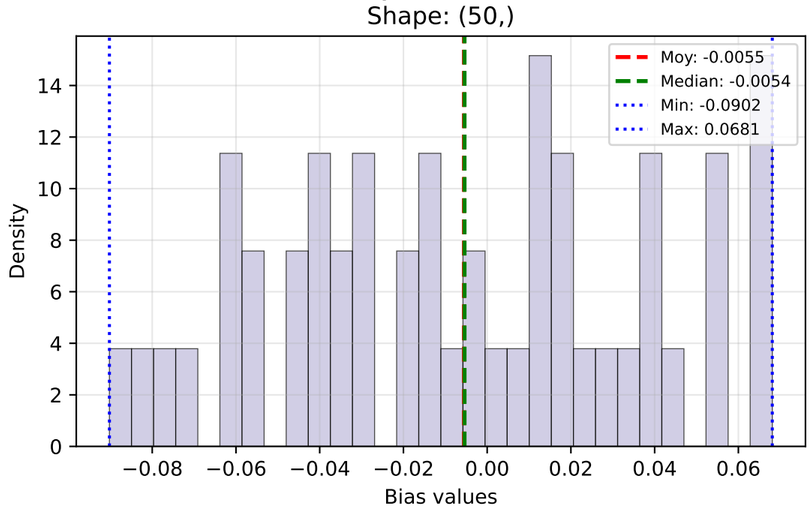
Layer 2 - WEIGHTS
Shape: (38, 50), Sparsity: 0.00%



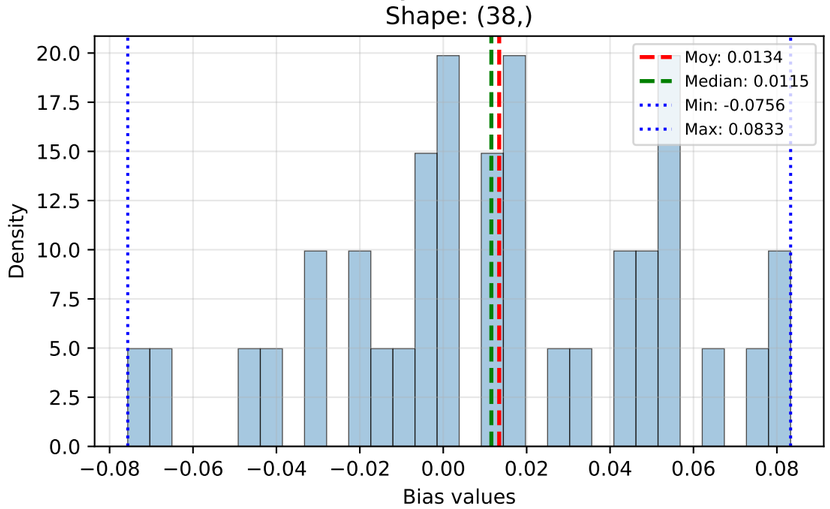
Layer 0 - BIAIS
Shape: (66,)



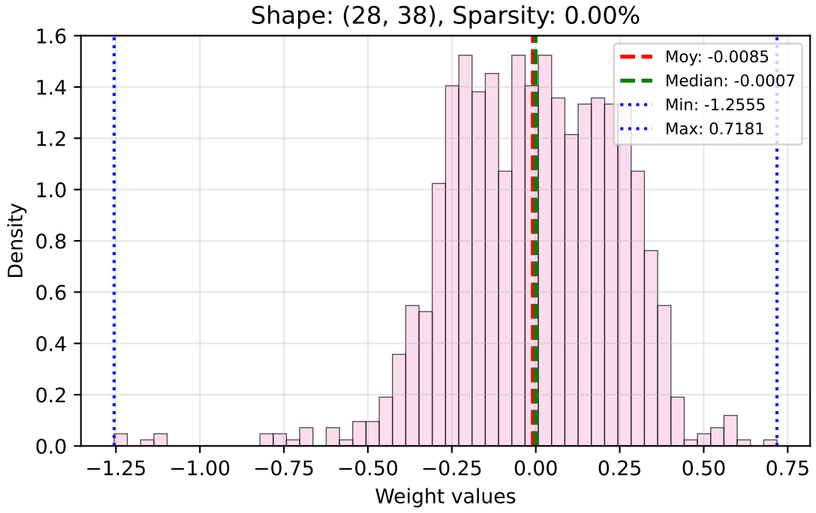
Layer 1 - BIAIS
Shape: (50,)



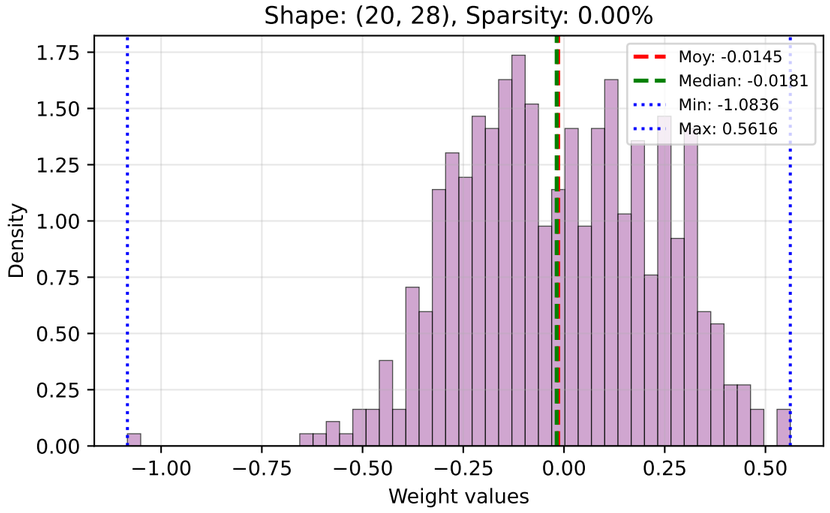
Layer 2 - BIAIS
Shape: (38,)



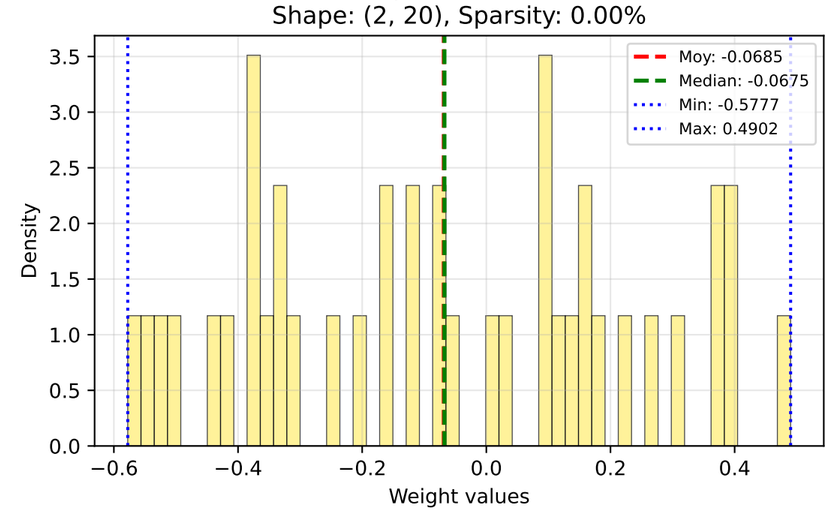
Layer 3 - WEIGHTS
Shape: (28, 38), Sparsity: 0.00%



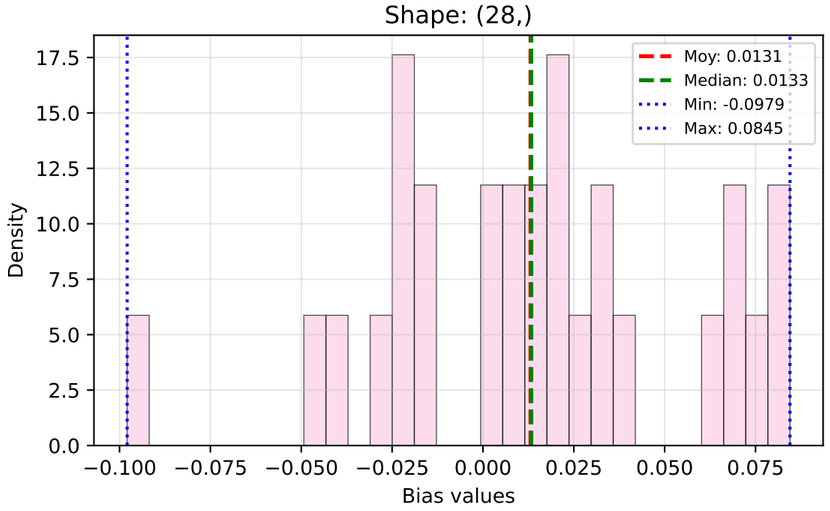
Layer 4 - WEIGHTS
Shape: (20, 28), Sparsity: 0.00%



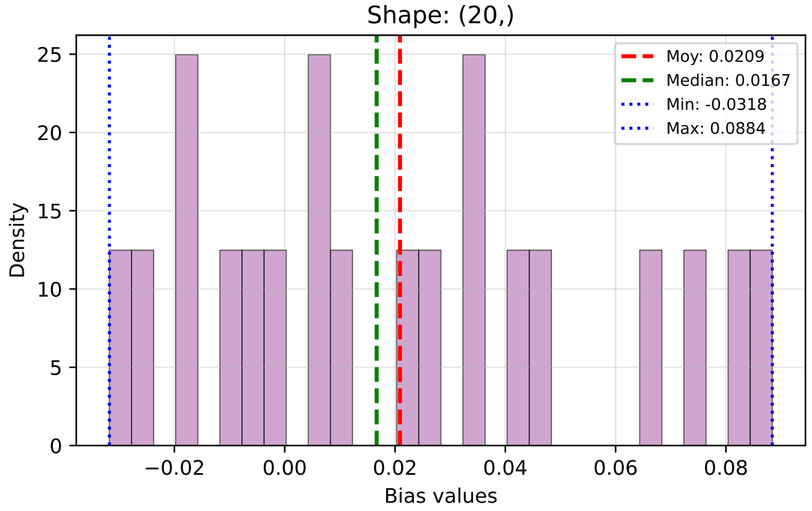
Output - WEIGHTS
Shape: (2, 20), Sparsity: 0.00%



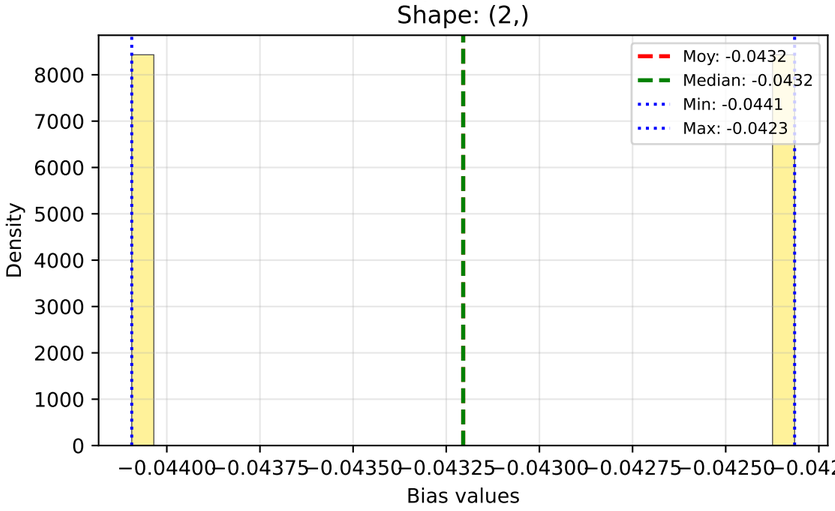
Layer 3 - BIAIS
Shape: (28,)



Layer 4 - BIAIS
Shape: (20,)

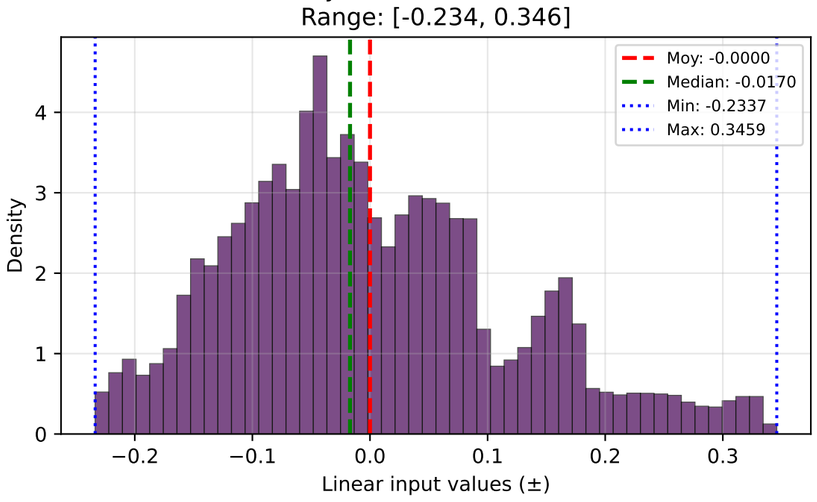


Output - BIAIS
Shape: (2,)

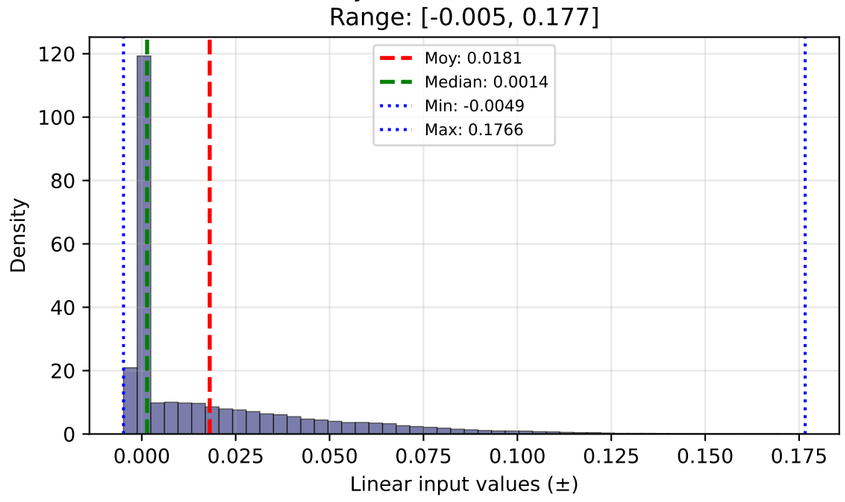


Activation distributions - Linear inputs/outputs and activations

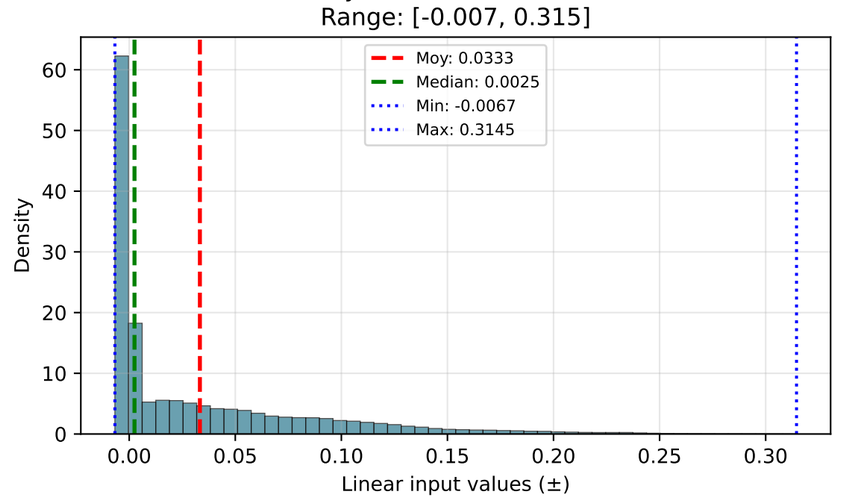
Layer 0 - LINEAR INPUT
Range: [-0.234, 0.346]



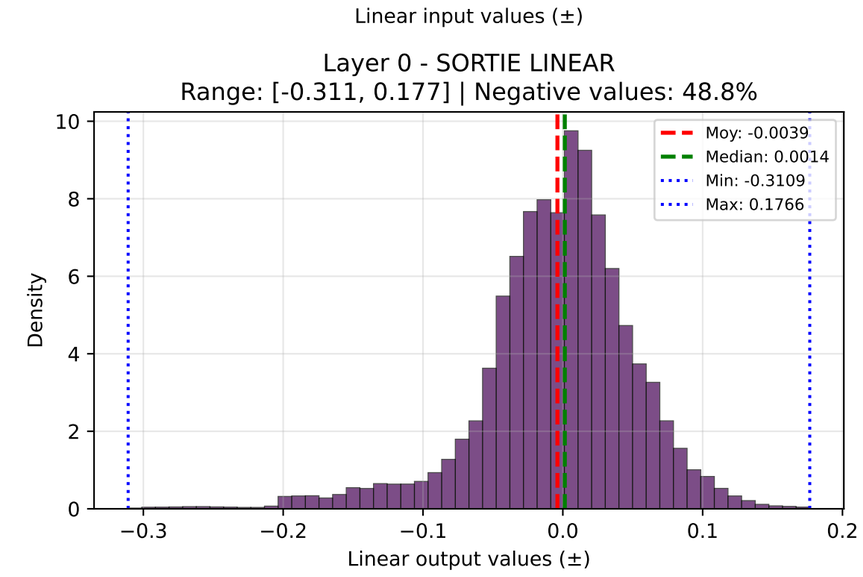
Layer 1 - LINEAR INPUT
Range: [-0.005, 0.177]



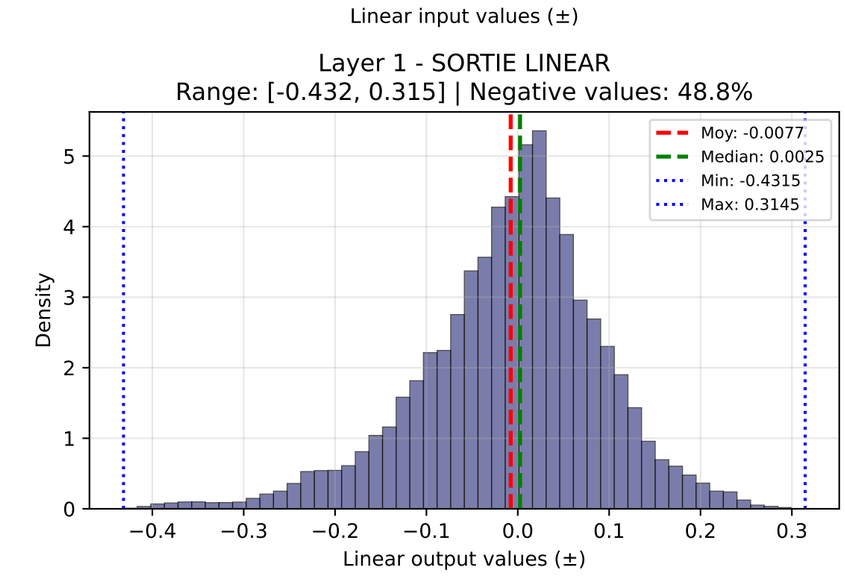
Layer 2 - LINEAR INPUT
Range: [-0.007, 0.315]



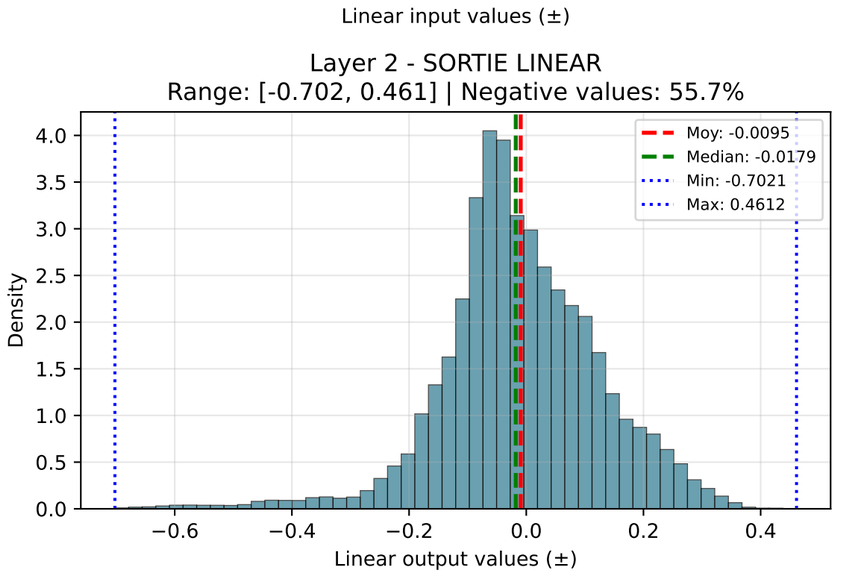
Layer 0 - SORTIE LINEAR
Range: [-0.311, 0.177] | Negative values: 48.8%



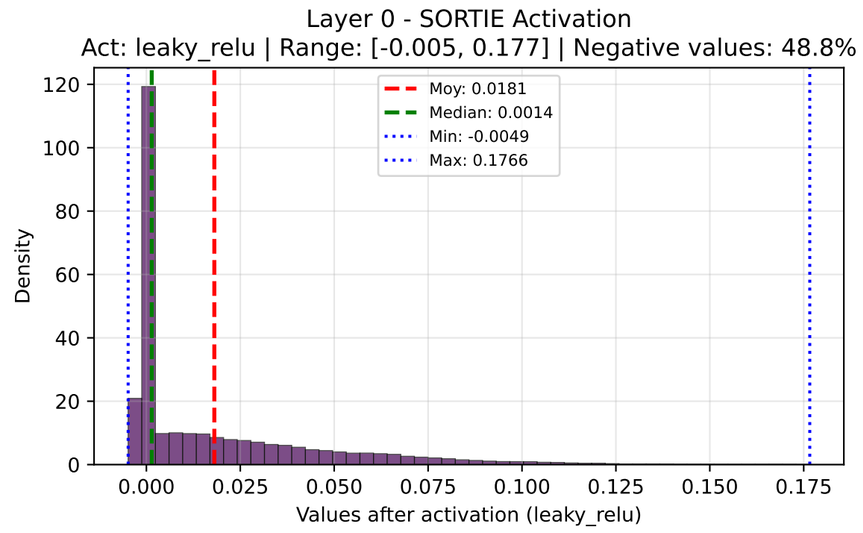
Layer 1 - SORTIE LINEAR
Range: [-0.432, 0.315] | Negative values: 48.8%



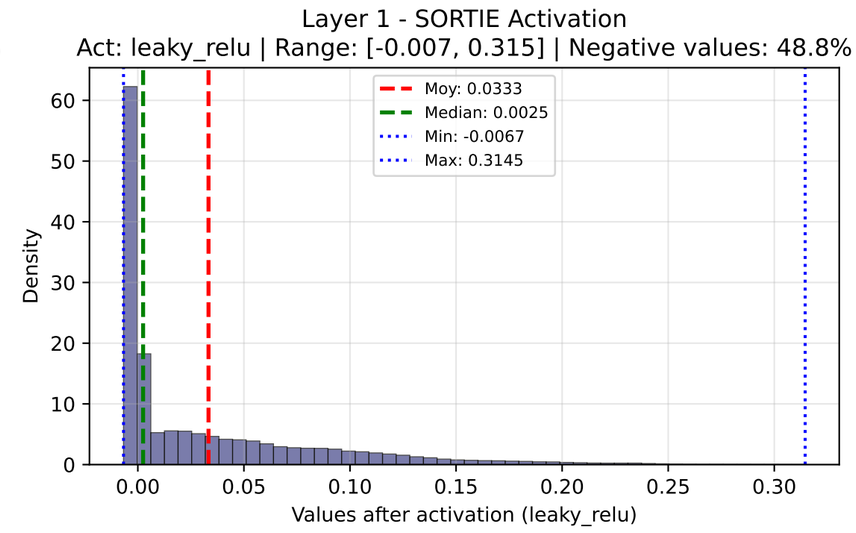
Layer 2 - SORTIE LINEAR
Range: [-0.702, 0.461] | Negative values: 55.7%



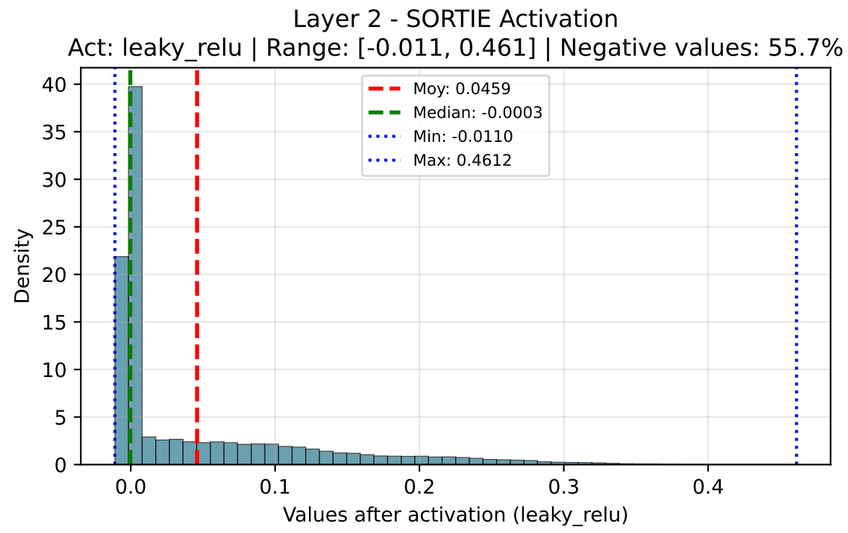
Layer 0 - SORTIE Activation
Act: leaky_relu | Range: [-0.005, 0.177] | Negative values: 48.8%



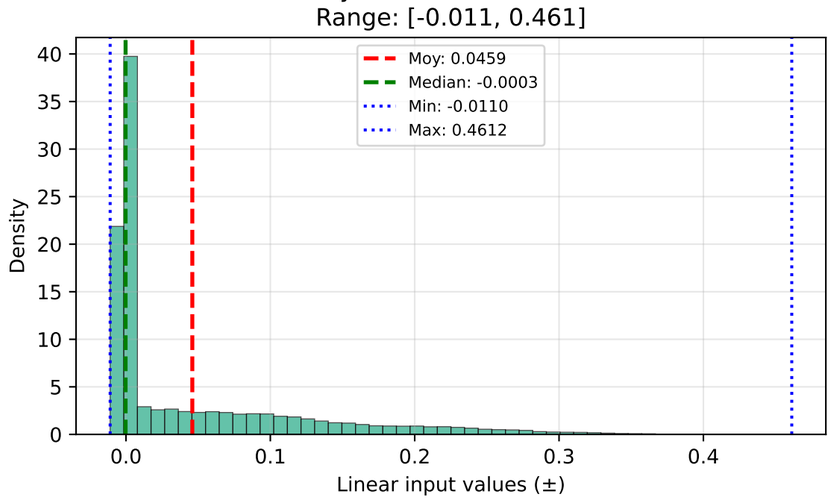
Layer 1 - SORTIE Activation
Act: leaky_relu | Range: [-0.007, 0.315] | Negative values: 48.8%



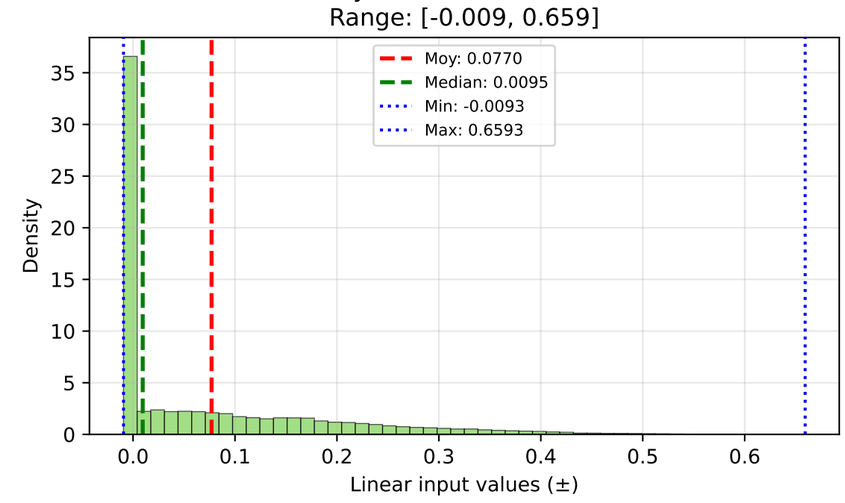
Layer 2 - SORTIE Activation
Act: leaky_relu | Range: [-0.011, 0.461] | Negative values: 55.7%



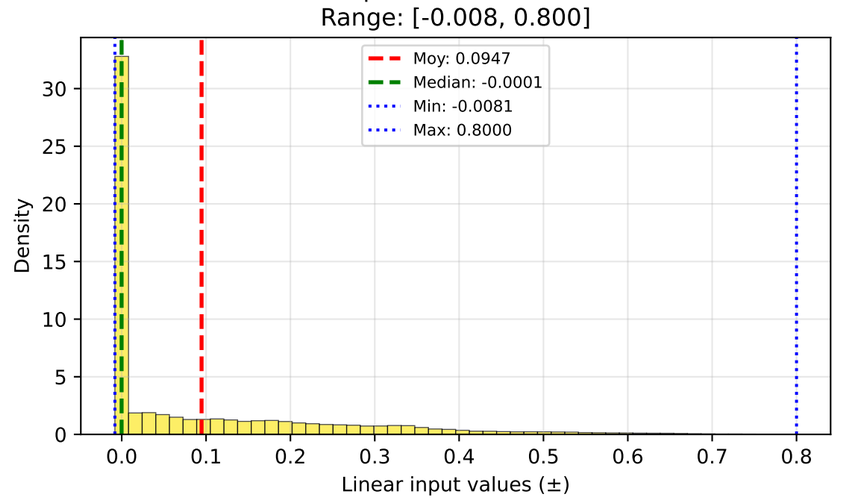
Layer 3 - LINEAR INPUT
Range: [-0.011, 0.461]



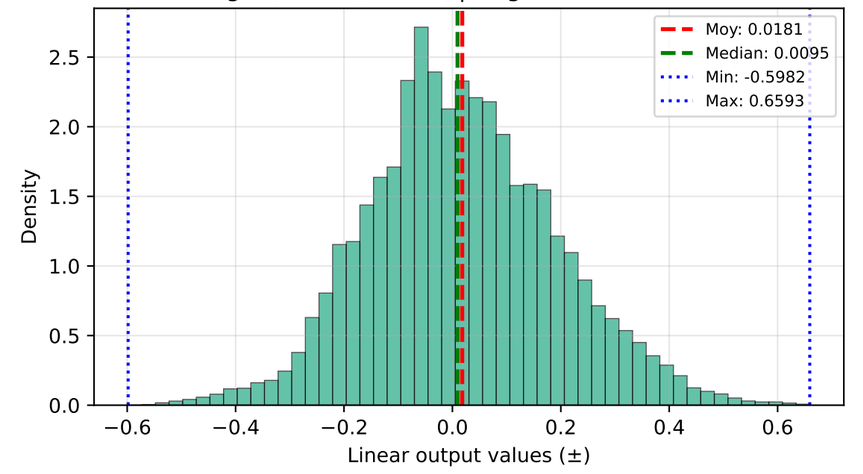
Layer 4 - LINEAR INPUT
Range: [-0.009, 0.659]



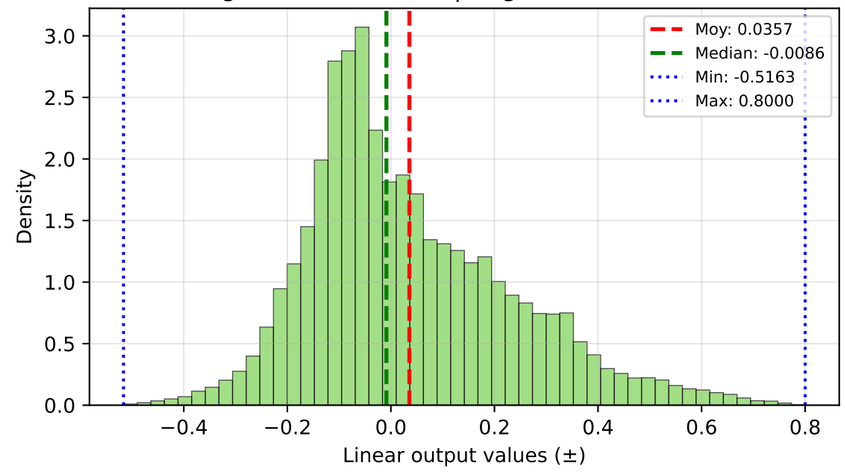
Output - LINEAR INPUT
Range: [-0.008, 0.800]



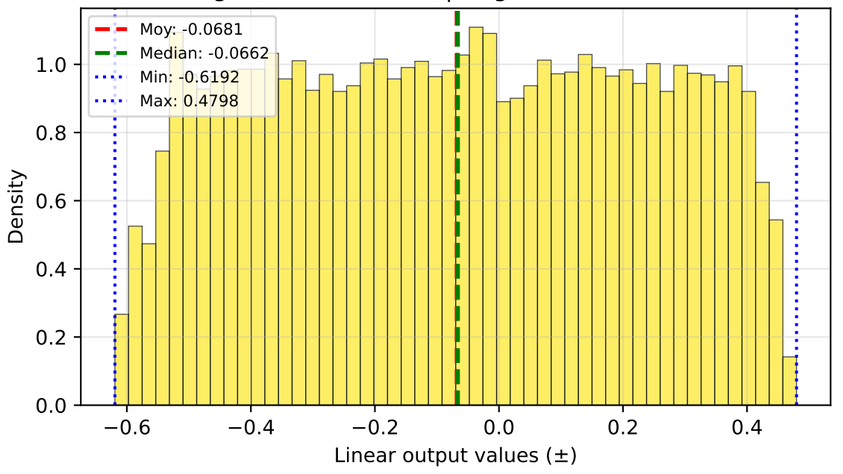
Layer 3 - SORTIE LINEAR
Range: [-0.598, 0.659] | Negative values: 48.2%



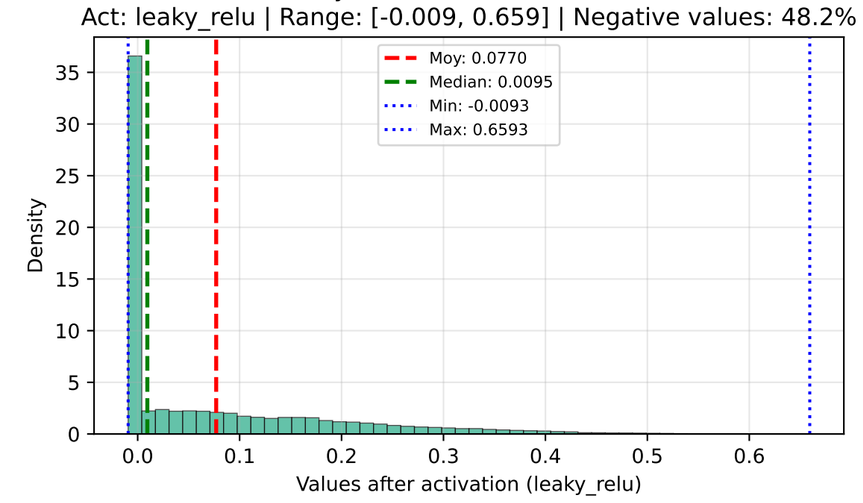
Layer 4 - SORTIE LINEAR
Range: [-0.516, 0.800] | Negative values: 51.5%



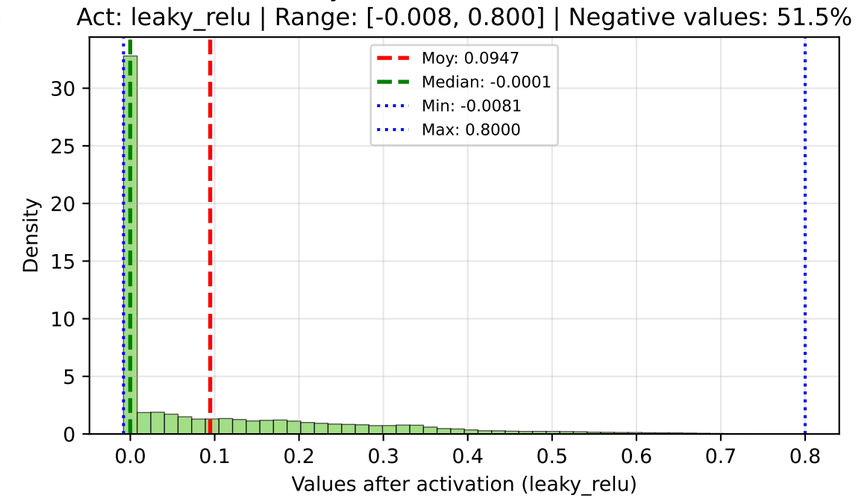
Output - SORTIE LINEAR
Range: [-0.619, 0.480] | Negative values: 57.1%



Layer 3 - SORTIE Activation
Act: leaky_relu | Range: [-0.009, 0.659] | Negative values: 48.2%



Layer 4 - SORTIE Activation
Act: leaky_relu | Range: [-0.008, 0.800] | Negative values: 51.5%



Output - NO ACTIVATION

No activation on the output layer

Quantization summary table

```
Architecture : (66, 50, 38, 28, 20)
Hidden activations : layer_0: leaky_relu | impl=analytic_int, layer_1: leaky_relu | impl=analytic_int, layer_2: leaky_relu | impl=analytic_int, layer_3: leaky_relu | impl=analytic_int, layer_4: leaky_relu | impl=analytic_int
Input dimension : 6
Output dimension : 2
Total number of parameters : 7,464
Dataset : dataset_walter.npz
V2 widths act/weights/bias/DSP : 16/16/16/48 bits
Calibration : mode=max, percentile=99.9
```

Tenseur	Layer	Kind	Activation	Impl.	N sel.	Shift exponent	Facteur scale	Bits	Float32 min	Float32 max	Int min	Int max	% Utilisation sym.
input	-	input	-	-	-	16	65536	16	-0.233695	0.345871	-15315	22667	69.2%
layer_0_weights	layer_0	weights	-	-	-	15	32768	16	-0.791346	0.617584	-25931	20237	79.1%
layer_0_bias	layer_0	bias	-	-	-	18	262144	16	-0.076065	0.057380	-19940	15042	60.9%
layer_0_preact	layer_0	preact	leaky_relu	analytic_int	-	16	65536	16	-0.310882	0.176613	-20374	11574	62.2%
layer_0_act	layer_0	act	leaky_relu	analytic_int	-	17	131072	16	-0.004858	0.176613	-637	23149	70.6%
layer_1_weights	layer_1	weights	-	-	-	14	16384	16	-1.216827	0.848078	-19936	13895	60.8%
layer_1_bias	layer_1	bias	-	-	-	18	262144	16	-0.090225	0.068105	-23652	17853	72.2%
layer_1_preact	layer_1	preact	leaky_relu	analytic_int	-	16	65536	16	-0.431540	0.314545	-28281	20614	86.3%
layer_1_act	layer_1	act	leaky_relu	analytic_int	-	16	65536	16	-0.006743	0.314545	-442	20614	62.9%
layer_2_weights	layer_2	weights	-	-	-	14	16384	16	-1.294705	0.848567	-21212	13903	64.7%
layer_2_bias	layer_2	bias	-	-	-	18	262144	16	-0.075649	0.083295	-19831	21835	66.6%
layer_2_preact	layer_2	preact	leaky_relu	analytic_int	-	15	32768	16	-0.702056	0.461222	-23005	15113	70.2%
layer_2_act	layer_2	act	leaky_relu	analytic_int	-	16	65536	16	-0.010970	0.461222	-719	30227	92.2%
layer_3_weights	layer_3	weights	-	-	-	14	16384	16	-1.255460	0.718122	-20569	11766	62.8%
layer_3_bias	layer_3	bias	-	-	-	18	262144	16	-0.097917	0.084488	-25668	22148	78.3%
layer_3_preact	layer_3	preact	leaky_relu	analytic_int	-	15	32768	16	-0.598237	0.659322	-19603	21605	65.9%
layer_3_act	layer_3	act	leaky_relu	analytic_int	-	15	32768	16	-0.009347	0.659322	-306	21605	65.9%
layer_4_weights	layer_4	weights	-	-	-	14	16384	16	-1.083596	0.561647	-17754	9202	54.2%
layer_4_bias	layer_4	bias	-	-	-	18	262144	16	-0.031776	0.088411	-8330	23176	70.7%
layer_4_preact	layer_4	preact	leaky_relu	analytic_int	-	15	32768	16	-0.516325	0.800009	-16919	26215	80.0%
layer_4_act	layer_4	act	leaky_relu	analytic_int	-	15	32768	16	-0.008068	0.800009	-264	26215	80.0%
layer_5_weights	layer_5	weights	-	-	-	15	32768	16	-0.577660	0.490198	-18929	16063	57.8%
layer_5_bias	layer_5	bias	-	-	-	19	524288	16	-0.044094	-0.042315	-23118	-22185	70.6%
layer_5_output	layer_5	output	-	-	-	15	32768	16	-0.619247	0.479789	-20291	15722	61.9%